

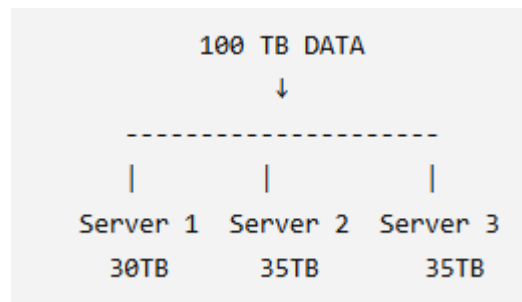
1. **Big Data** refers to very large, fast-growing, and diverse amounts of data that cannot be efficiently stored, processed, or analyzed using traditional single-machine systems.

The problem is:

size, speed, complexity, or diversity creates challenges for traditional systems.

We cannot efficiently handle the entire workload (large data) on one machine.

So we use **multiple machines** to store and process the data.

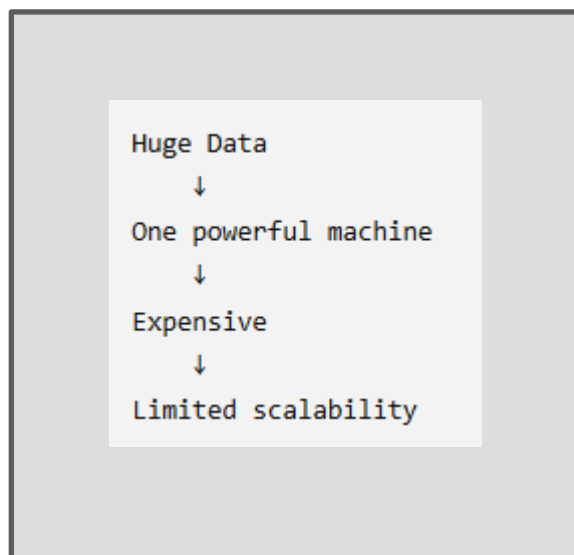


storage and processing are distributed.

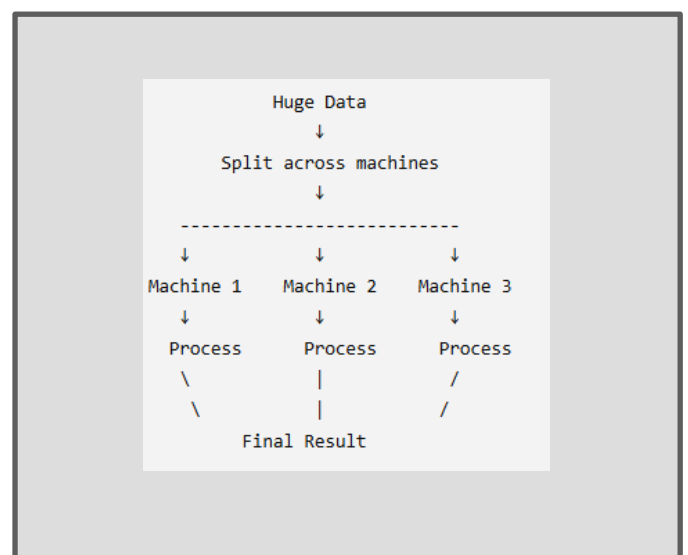
Advantages

1. More storage
2. More processing power
3. Parallel processing
4. Better scalability
5. Fault tolerance

Traditional approach



Big Data approach



NOTE:

Distributed Computing : Instead of buying one extremely powerful machine, use multiple machines and divide the work among them.

2. 5Vs of Big Data (Vs are characteristics/challenges of Big Data)

V1 — Volume = Amount of data

V2 — Velocity = Speed at which data is generated, received, and processed.

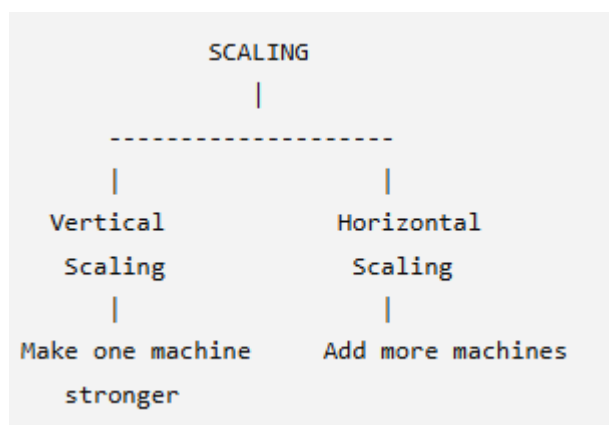
V3 — Variety = Different types/formats of data (Structured,unstructured,semi-Structured)

V4 — Veracity =Trustworthiness/quality of data.(Can we trust the data? Age = 25, Age = NULL, Age = -10)

V5 — Value = Useful information/business benefit obtained from the data.

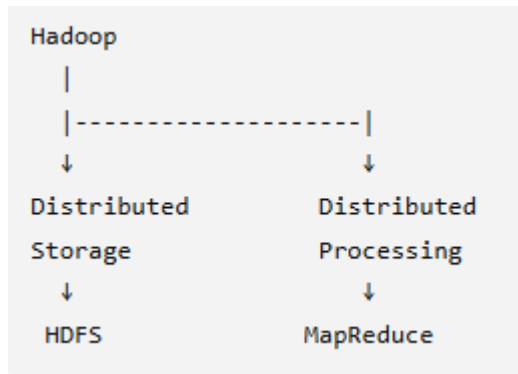
V	Meaning	Simple question
Volume	Amount	How much data?
Velocity	Speed	How fast is data coming?
Variety	Types	What kinds of data?
Veracity	Quality	Can we trust it?
Value	Benefit	Is it useful?

3. Scaling means increasing or decreasing the computing resources of a system according to the workload.

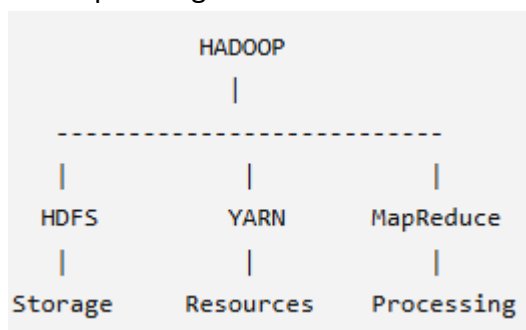


4. Distributed computing is a computing approach where a large task is divided into smaller tasks and executed across multiple machines.

5. Parallel processing means all machines can work simultaneously.
6. Apache Hadoop is an open-source framework used for distributed storage and distributed processing of large datasets across clusters of computers.

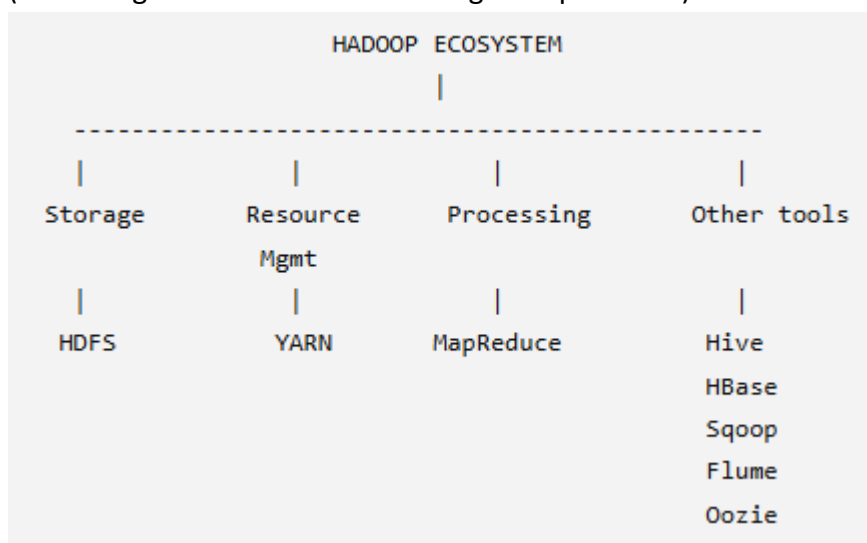


Hadoop manages cluster resources



Hadoop Ecosystem

(technologies that solve different Big Data problems)



Component	Main purpose
HDFS	Distributed storage
YARN	Resource management
MapReduce	Distributed processing